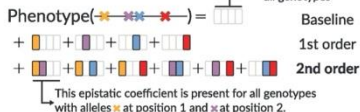


Linear Epistatic coefficients

Genotype with L=4 variable sites: 

Epistatic coefficient decomposition at order 2:



To infer epistasis coefficients, one first need to choose the **framework of the model**. This framework choice **does not modify the accuracy of the model** but changes the interpretation of the coefficients.

Common frameworks

- RBA  = 0
Coefficients with a reference allele are set to zero.

Coefficients represent epistatic effects relative to a **reference genotype**.



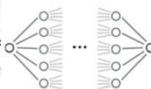
The two sets of coefficients are **linearly related**

- RFA  = 0
The sum of all alleles at a specific site is 0

Coefficients represent epistatic effects relative to the **average genotype of the dataset**.

Interpretations

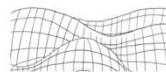
RBA coefficients are useful when there is a reference (known ancestor or wildtype sequence). They can be used to infer the likelihood of directional evolutionary paths or to measure biological interactions between alleles.



These coefficients depend on the maximum order of the epistatic model inferred, we recommend a train-test approach to select the best order.

 Inferred at order 1 is different from  Inferred at order 2

RFA coefficients can be computed independently for each order. Hence, they can be used to infer the fraction of phenotypic variance attributable to epistasis at each order. It's one way to quantify the complexity, or "ruggedness" of a landscape.



These coefficients depend on the specific variable sites considered.

 Adding new sites can drastically modify existing coefficients.

Inference methods

Hadamard method:

- RFA framework only
- Fast and memory efficient
- Does not work with missing data

Linear regression:  **Recommended method in most cases**

- All frameworks
- Works with missing data
- Compatible with non-linear effects

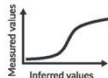
"Exact" / naive method:

- RBA framework only
- Easy to explain
- Inconsistent estimator, totally unreliable when data is noisy 
- Does not work with missing data

Adding non-linear effects

Non-linear effects generally break the equivalence between frameworks.

Global epistasis Phenotypes are projected onto an additive scale by a nonlinear transformation.



This transformation can be performed prior to inferring coefficients, or jointly.

Sparsity constraints

L0 norm, lasso, ridge regression, etc. Push small coefficients towards 0.

Phenotypes can require a global transformation to account for a nonlinear relationship between them and an underlying, additive, phenotype. Use with caution, as they distort specific epistatic interactions.

Should I use global epistasis ?

Complex phenotypes (fitness / cell size...)



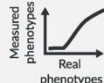
Yes! But test with and without.

Phenotypes expected to be additive (energies...)



Maybe, is there a theoretical justification?

Measurement limits



No, prefer calibration curves and censored statistics.

Sparsity constraints can help fix neutral mutations coefficients to 0 and avoid overfitting noise, but can also erase real, low-level, background epistasis.

Linear epistatic models can fit any landscape at high enough order. Follow proper statistical practices to avoid overfitting (cross-validation, model comparison, bootstrap).